# 8. The Functional Anatomy of Enterprise AI Demand

## A seven-function framework for modeling B2B token consumption, 2026–2028

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# 1. Executive Summary

Enterprise demand for AI inference is among the fastest-growing line items in corporate technology budgets, and the input that ultimately sets the slope of global GPU and memory-semiconductor demand. This report maps that demand across **seven business functions** — classified by what each task computationally is rather than by industry sector — and quantifies how token consumption is distributed across them and how it evolves through 2028. This report deliberately stops at the token layer: it produces the demand function, stated in tokens and in compute-weighted (frontier-equivalent) tokens, and leaves the conversion of that function into hardware requirements to the companion infrastructure volume.

The organizing insight is that two properties of a task jointly determine both how fast agentic AI will transform it and how many tokens that transformation will consume. The first is the **verification mechanism**: who or what judges the output's correctness, how quickly, and at what cost. Tasks with deterministic, machine-checkable verification (a compiler, a test suite, a reconciliation that either ties or does not) permit autonomous retry loops and therefore explosive per-task token amplification. Tasks verified only by serial human reading remain throttled by human attention — and the throttle binds hardest where the effort ratio inverts: an agent can work unattended for days on code that a test suite verifies in minutes, but thirty minutes of qualitative agent work can produce a document that consumes a full day of careful reading. The second is the **execution harness**: autonomous sandbox loops, parallel batch pipelines, real-time streaming, or connector orchestration across enterprise systems — each with a distinct token profile.

Applying this framework, we partition B2B work into: (1) Software Engineering; (2) Quantitative Analysis & Data Science; (3) Transaction Record Operations; (4) Sales & Outreach; (5) Customer Interaction; (6) Qualitative Text Review & Deep Research; and (7) Business Workflow Integration & Automation. An eighth function in the full taxonomy — Generative Media — runs on a separate compute axis because its unit of account (visual and audio generation) is not convertible to LLM tokens; its demand structure is developed in the companion volume on commercially produced generative media. The seven functions are a deliberately narrow slice of enterprise AI rather than the whole of it: chatbot-tier knowledge work — by volume the largest block of enterprise usage — is excluded by construction, as are several agentic domains for which no defensible base population yet exists. Section 2.3 states the scope precisely, and the figures throughout should be read as a floor on the seven functions named.

Our quantitative model produces three headline results. First, **B2B LLM token demand across the seven functions grows from roughly 20,100 trillion tokens in 2026 to 38,100 trillion in 2028** — a 1.9x expansion — with software engineering's share declining from 82% to 57%. The shape of that decline matters more than its size: 2026 is the year in which agentic coding's growth peaks — the intensity-normalized penetration of the developer base jumps to 55% — and coding's year-over-year growth then decelerates to 1.18x in 2027 and 1.11x in 2028, while every other function grows roughly three- to ten-fold. Coding does not retreat; it stops being the growth engine. Second, the post-coding growth is led by Quantitative Analysis & Data Science, which quadruples to roughly 8,400 trillion tokens by 2028 — the one non-coding knowledge-work function that pairs a machine-checkable feedback loop (code executes; the diagnostics are numbers) with an autonomous-loop harness, and the one whose felt utility to

non-experts is largest — and therefore the fastest-diffusing: it rises immediately through tool-embedded agents (Excel and notebook add-ins operating on files the user already holds, a path the enterprise semantic layer does not gate) and accelerates as the H2-2026 semantic-layer build-out widens the self-serve analyst base. Business Workflow Integration & Automation grows eight-fold to roughly 3,200 trillion as that same build-out connects the systems agents act across. Qualitative Text Review & Deep Research reaches roughly 3,000 trillion tokens by 2028 and begins the forecast with the highest penetration of any non-coding function, 18% — it is the easiest agentic capability to reach — but grows only 2.9x, because it remains the mature, human-verified workload: its final artifacts must still pass through a human reading at roughly 200–300 words per minute, and that review bandwidth, not agent capacity, caps commissioning frequency. The ordering of growth rates across the seven functions largely follows their verification mechanisms: functions verified by a machine or by a statistical outcome grow four- to ten-fold, the one verified by a human reading grows 2.9x, and the one that had already saturated its verifier grows 1.3x. The instructive exception is workflow automation, which grows eight-fold with no verifier at all at the point of action — catch-up from a very low base as the control plane it depends on is built, rather than autonomy. Third, **token share is not compute share**. Because low-wage, structured tasks route to budget models whose per-token FLOP cost is five to ten times lower than frontier models, our compute-weighted demand — the series that actually maps to infrastructure requirements — runs roughly 32–38% below the naive token total, growing from approximately 13,700 trillion frontier-equivalent tokens in 2026 to 23,700 trillion in 2028. Under the category-level compute weights this model assumes, an analysis that converts nominal token counts to hardware with a single coefficient would overstate requirements by that margin — a scenario result that follows from the assumed weights, not an empirically measured error band.

The strategic implications follow the taxonomy. The seven functions divide into distinct workload profiles — long-context autonomous loops, latency-tolerant parallel batches, real-time streaming, connector orchestration — whose divergent silicon consequences the companion infrastructure volume develops. Within the scope of this report, the central finding is that the **semantic layer** — the metadata and ontology infrastructure that gates the second wave of Categories 2 and 7 — is a control-plane asset in the precise sense developed in our earlier work: whoever owns it owns the gateway to the next major tranche of token demand.

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# 2. Introduction

## 2.1 The 2026 inflection and the unit of analysis

The empirical starting point of this report is the most violent demand acceleration enterprise software has ever recorded. Anthropic's annualized run-rate revenue rose from roughly $1 billion in January 2025 to $9 billion at year-end, then advanced through $14 billion in February 2026, $19 billion in March, and $30 billion in April, crossing $47 billion by mid-May — disclosed by the company alongside its $65 billion Series H. Three qualifications discipline that headline. It is a run rate, the most recent month annualized, not audited revenue: full-year 2026 revenue on an accrual basis will land far below it. It is reported gross, counting

end-customer spend routed through cloud resellers as revenue and booking partner payouts as expense, which inflates comparison with net-reporting peers. And $47 billion remains the last company-confirmed figure as of this writing; third-party trackers place the late-July run rate materially higher, but those trackers have historically run hot and the number is unverified. The customer-side series is the more informative one for demand analysis, being less sensitive to annualization: more than 1,000 enterprise customers each spending over $1 million annually, a count that doubled in under two months; eight of the Fortune 10; and roughly 80% of revenue from business customers — enterprise API and developer contracts rather than consumer subscriptions. Anthropic's confidential IPO filing on June 1, 2026 will, when it becomes public, subject this trajectory to audited disclosure for the first time. The proximate driver is agentic coding: Claude Code passed a $2.5 billion run rate within nine months of launch, the last figure the company itself disclosed, with third-party estimates placing it several times higher by mid-2026; and on OpenRouter — the largest multi-model routing platform, reportedly processing on the order of 100 trillion tokens per month, a five-fold increase in six months — programming workloads grew from roughly 11% of total token consumption in early 2025 to above half by late 2025, a share still reported through the first quarter of 2026.

The question this report addresses is the one any infrastructure investor must now ask: as coding's growth rate inevitably moderates from its 2025–26 hypergrowth (a dollar-growth deceleration our companion report, *The Token Economics of Coding Agents*, analyzes in depth and characterizes as revenue-quality normalization rather than demand destruction), **which business functions inherit the growth, in what sequence, and at what token intensity?**

This report organizes the analysis around business functions rather than industry sectors, for a reason best seen through a single example. Consider three workers inside the same financial institution: a developer building a fintech application, a quantitative analyst backtesting strategies against market data, and a research analyst reviewing a five-hundred-page sell-side report. All three sit in the "financial services AI" segment of a vertical taxonomy. Yet their workloads have almost nothing in common. The developer's agent runs autonomous compile-test-fix loops whose token consumption compounds across dozens of iterations; the quant's agent executes code against statistical verification metrics; the research analyst's agent produces text whose only verifier is the analyst's own serial reading. Their token volumes differ by one to two orders of magnitude per task, and their model-tier requirements differ — frontier reasoning for judgment-heavy review, budget models for structured processing.

Conversely, the developer in that bank runs a workload structurally identical to a developer in a manufacturer, a retailer, or a software firm. The sector tells us *who pays*; it tells us almost nothing about *what computes*. Since the purpose of this report is to aggregate workloads into a demand function, we organize the analysis around business functions — the unit at which token mechanics are homogeneous — and treat sectors merely as weighting vectors over those functions.

A further consideration is that the occupational statistics most often cited in this area — including Anthropic's own Economic Index, which classifies usage against O*NET occupation codes — are themselves beginning to blur. Anthropic's finance agent templates run as plugins inside Claude Code and Claude Cowork; OpenAI explicitly positions Codex as the mechanism for carrying agents into enterprise domains beyond coding. A reconciliation job executed by

a financial analyst through a coding harness registers, statistically, as "Computer and Mathematical" work. Functional classification by verification mechanism and harness type is robust to this drift; occupational classification is not.

## 2.2 What the model is and is not

The quantitative model presented in Section 5 is deliberately coarse: a single multiplicative identity per category (base population or volume × activity rate × tokens per task × penetration), with one compute weight per category, and no prefill/decode decomposition. Every coefficient is an explicit, adjustable assumption, and the model is built to deliver order-of-magnitude conclusions about mix and trajectory rather than point estimates. Section 6 documents the sensitivity of each input and the confidence attached to it.

## 2.3 Scope of this estimate

The demand modeled here is not the whole of enterprise AI. It is a deliberately narrow slice, and the exclusions are larger than the inclusions.

We exclude chatbot-tier knowledge work entirely — background research, email drafting, summarization, proofreading, translation. By volume of interactions this is almost certainly the largest single block of enterprise AI usage today, and it is growing. It is excluded not because it is unimportant but because it is a different object: a single-turn request answered from a single model call, with no tool loop, no autonomous execution, and no verification structure of the kind this report is built around. For the same reason we exclude what employees do on personal free-tier and consumer accounts, whether or not the work is for their employer: it is chatbot-tier usage by another route, and it appears in no paid-seat base.

Within agentic work, the seven functions modeled here were selected on one criterion: each has a verification mechanism that can be named and reasoned about. That criterion selects for tractability, not for size. Enterprise R&D — computational chemistry, materials discovery, simulation-in-the-loop engineering, scientific literature work — is agentic, is verification-rich, and may in time exceed several of the functions modeled here. It is absent because no defensible base population for it can yet be constructed. The same applies to industrial process optimization and supply-chain planning.

Demand arising inside China is likewise outside the estimate, though the exclusion operates differently across the model and one route is weaker than the others. The non-coding workforce categories are built on Microsoft 365 commercial seats, a base from which China-domiciled demand is largely absent by construction. The two volume-based categories do not run on seat bases at all — Category 3 is sized on addressable transaction items and Category 5 on customer interactions — and both are defined here on an ex-China basis. Category 1 is the exception: its base is a published global developer population from which China-domiciled developers cannot be separately identified, so the exclusion there operates only through the intensity-normalized penetration term, which measures the share of that population operating at the reference intensity of this report's frontier-class agentic tooling. Readers should treat Category 1's base as global and its penetration term as carrying the geographic adjustment, as Section 4.1 sets out.

The figures that follow should therefore be read as a floor on the seven functions named, not as an estimate of enterprise AI demand. For the same reason, they are not expressed anywhere

as a share of installed accelerator capacity: that comparison would set a narrowly scoped floor against an all-purpose stock. Where scale needs conveying, the companion infrastructure volume does so in gigawatts, capital expenditure, and annual increments.

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## 3. Verification × Harness

### 3.1 The economics of verification cost

The foundational law of agentic AI economics is that **the value created by generation is gated by the cost and speed of verification.** A model can produce a plausible artifact — code, a reconciliation, a legal memo — in seconds; the binding constraint is establishing that the artifact is correct. The rigorous empirical demonstration comes from the software domain. Tracking more than 100,000 GitHub developers, the NBER event study of Demirer, Musolff, and Yang (2026) found that AI coding tools raised commit output by a cumulative 40–180% across successive tool generations while release-level output rose only about 30% — a pattern the authors show is consistent with a weak-link hypothesis in which human bottlenecks, verification chief among them, constrain the translation of code output into shipped releases, with an estimated elasticity of substitution between AI and human effort of just 0.25. The entire subsequent history of coding agents, as our companion report documents, is the history of automating that verification away.

Verification mechanisms fall into six classes, and the classification of a task here predicts its transformation speed better than any other single variable:

**Deterministic machine verification.** A compiler, a test suite, a record match. Verdicts are instant, free, and binary, so an agent can retry autonomously until success. This is the regime of exponential productivity scaling and compounding token consumption.

**Semi-deterministic verification.** Code executes and statistical metrics (fit diagnostics, back-test performance, p-values) provide quantitative feedback, but a residual human judgment — is this model specification sensible? — closes the loop. Autonomous iteration works; the loop is slightly shorter.

**Outcome-based verification.** Customer-service resolution rates, email reply rates. Correctness is measured statistically over many trials rather than per instance. Automation proceeds, but improvement cycles run at the cadence of A/B measurement, not compile cycles.

**Sensory-parallel human verification.** The human visual cortex evaluates an image or video draft in roughly half a second. Iteration is rapid but a human remains in every loop. (In the B2B taxonomy this regime belongs almost entirely to generative media, treated in the companion consumer volume.)

**Cognitive-serial human verification.** Legal memos, research reports, strategy documents can be validated only by reading — roughly 200–300 words per minute, line by line. This is the most expensive verification regime, and the one where hallucination risk imposes the steepest tax. It also inverts the effort ratio that lets the machine-verified regimes scale. In coding or quantitative analysis, an agent can work unattended for days while the human verifies the result in minutes — the test suite goes green, or the deliverable is a handful of tables and charts

absorbed at a glance. In cognitive-serial work the ratio runs the other way: thirty minutes of agent effort can produce a document that consumes a full day of careful reading. Where verification is machine-checked, AI effort *substitutes for* human effort; where it is cognitive-serial, AI effort *generates* human effort — and a rational professional therefore rations the work they commission, however cheap each run becomes.

**Unverified side-effectful execution.** Sending an email, approving a payment, mutating a record: the action is irreversible, and no oracle — machine or human — can confirm its correctness before the consequences land. Verification here is not slow or expensive; it is structurally absent at the moment it is needed. Autonomous retry loops therefore cannot close, human-approval gates persist at the point of action, and the regime is confined to augmentation rather than full automation regardless of model capability. (In the B2B taxonomy this regime is occupied by cross-system workflow execution, Category 7.)

## 3.2 Execution harnesses and token amplification

The second axis determines how many tokens a unit of work consumes. Four harness archetypes recur. **Autonomous loops** (terminal/sandbox agents) re-inject growing context on every iteration; a single agentic coding task has been reported to consume up to 20 million tokens, against roughly 30 thousand for a simple chat summary — a nearly three-orders-of-magnitude gap. Token accumulation across a session is superlinear (approximately quadratic in turns, moderated by prompt caching of the append-only prefix). **Parallel batch** pipelines run thousands of independent agent instances with no latency constraint — one researcher per lead, one processor per claim. Per-instance tokens are moderate; aggregate volume is set by instance count, which scales with business volume rather than headcount. **Real-time streaming** (voice agents) consumes modest tokens per interaction but binds infrastructure through latency SLAs. **Connector orchestration** chains calls across enterprise systems through text APIs, MCP servers, and CLIs; per-step tokens are moderate and cache well, but the binding constraint is the absence of a deterministic verifier for irreversible actions rather than raw compute. (We note that screenshot-driven computer-use agents, though frequently demonstrated, appear far less prevalent in production than connector-based execution, constrained by latency, cost, reliability, and security; text connectors are the workload that predominantly executes.)

The seven categories of Section 4 occupy seven economically significant cells of the verification × harness matrix; not every combination is populated, and several are empty or negligible. The matrix also predicts, critically for forecasting, which categories can follow coding up the amplification curve. Tasks with machine-checkable verification are the ones positioned to graduate to autonomous loops; the rest scale by volume or remain human-throttled.

## 3.3 Token share is not compute share

A third element completes the framework. The Anthropic Economic Index documents that model-tier selection correlates with task wage: each $10 increment in a task's associated hourly wage raises the share of Opus-class (frontier) model usage by 1.5 percentage points on Claude.ai and 2.8 points in the API. Inverted, this says that low-wage, structured tasks route to budget models — and on OpenRouter, the token-share leaders in agentic and batch workloads are precisely the Chinese budget models (MiniMax, GLM, Kimi, DeepSeek), which in some weeks of early 2026 accounted for roughly 60% of top-ten token consumption. Because a light-

weight mixture-of-experts model activates a fraction of the parameters of a frontier model, the same nominal token can plausibly differ by 5–10x in FLOPs — an architectural inference this model adopts as an assumption, not a per-model measurement. Any analysis that converts token counts to infrastructure demand — or, for that matter, to vendor revenue — with a single coefficient therefore overstates wherever cheap-model-sufficient categories grow. We handle this with a per-category **compute weight** (frontier = 1.0), and report both nominal and weighted totals throughout, with the weighted series expressed in frontier-equivalent tokens. Token share, revenue share, and compute share are three different distributions over the same taxonomy, and analysts tracking "AI usage" through any single metric will misread the market in proportion to that metric's distance from the question they are actually asking. The conversion of the weighted series into hardware requirements is the subject of the companion infrastructure volume.

### 3.4 The legacy-automation baseline

Finally, a point that materially shapes the sizing: several functions were substantially automated *before* generative AI, by RPA, rules engines, OCR pipelines, and template-driven marketing platforms. Structured invoice matching, standard claims adjudication, and mass email campaigns did not await LLMs. The LLM-addressable base in those functions is therefore not the gross volume of work but the **unstructured, exception-handling residual** that legacy automation could not reach — a distinction that holds the addressable volume for Categories 3 and 4 to a fraction of the gross, and one that most market sizing in this space overlooks.

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## 4. The Seven Functional Categories

Each category below follows the same four-part structure: (a) definition and boundaries; (b) work characteristics and verification mechanism; (c) agentic AI deployment today; (d) the 2026–2028 trajectory and the assumptions our model encodes. Table 1 summarizes. One scope note: the full verification-×-harness taxonomy contains an eighth function, Generative Media, which we exclude here not as a judgment on its importance but because its unit of account is not convertible to LLM tokens, its supplier ecosystem is disjoint from the LLM stack, and its demand is dominated by consumer-facing monetization pools; the companion consumer volume develops it in full.

**Table 1. The seven B2B functional categories**

| # | Category | Verification | Harness | Tokens/task (K) | Penetration 26→28E | Compute weight |
|---|---|---|---|---|---|---|
| 1 | Software Engineering | Deterministic | Autonomous loop | 500 | 55% → 72% | 0.70 |
| 2 | Quantitative Analysis & Data Science | Semi-deterministic | Autonomous loop (code exec) | 250 | 10% → 42% | 0.60 |

| # | Category | Verification | Harness | Tokens/task (K) | Penetration 26→28E | Compute weight |
|---|---|---|---|---|---|---|
| 3 | Transaction Record Operations | Deterministic | Parallel batch | 40 | 3% → 22% | 0.10 |
| 4 | Sales & Outreach | Outcome-based | Parallel batch | 100 | 2% → 20% | 0.15 |
| 5 | Customer Interaction | Outcome-based | Real-time streaming | 20 | 3% → 28% | 0.20 |
| 6 | Qualitative Text Review & Deep Research | Cognitive-serial | Parallel batch (deep-research runs) | 400 → 600 | 18% → 35% | 0.80 |
| 7 | Business Workflow Integration & Automation | Unverified side-effectful | Connector orchestration (text API/MCP/CLI) | 60 | 3% → 24% | 0.25 |

### 4.1 Software Engineering — the anchor workload

**Definition and boundaries.** All work whose deliverable is functioning code: feature development, bug fixing, test authoring, infrastructure-as-code, ETL pipeline construction, legacy migration — and, importantly, database engineering (schema design, query optimization), which builds the systems other categories run on. The base is roughly 40 million software-engineering developers — the upper band of published developer-population estimates, for a reason developed below. Coding-capable data scientists belong to Category 2, not here: although their harness and adoption timing resemble developers', their deliverable is a numerical answer or model rather than reusable code, and the boundary between the two categories is the deliverable, not the toolchain. This keeps the category a clean map onto the software-engineering installed base — GitHub's ~150 million registered developers as the outer TAM, GitHub Copilot's ~4.7 million paid subscribers as the observed monetization anchor.

**Why the upper band of the population estimates.** Published counts of the world's developers range from roughly 30 million professional software engineers to roughly 47 million once part-time, adjacent and hobbyist coders are included. Before agentic coding, the narrower figure was the operative one: a part-time coder could not consume tokens at anything like a professional's rate, because they could not produce code at anything like a professional's rate. Agentic coding removes that constraint. An analyst, a designer, or an operations engineer directing a coding agent generates tokens in proportion to the work delegated, not to their own fluency, and the population that consumes coding tokens is now defined by who delegates code, not by who holds the job title. The denominator therefore moves toward the upper band. We adopt 40 million rather than the full 47 million because the intensity-normalized penetration rate below already absorbs the lower average intensity of the newly included population.

**Work characteristics and verification.** This is the canonical deterministic-verification, autonomous-loop category. The compiler-and-test-suite stack provides free, instant, binary verdicts; the agent loops until green. The result is the most extreme token amplification in

the economy and — per the Economic Index — a visible migration of the workload itself from interactive consumer surfaces to automated API pipelines: between August 2025 and February 2026, the category's task share rose 14% in first-party API traffic while falling 18% on Claude.ai, the clearest statistical signature of work crossing from augmentation into automation.

**Deployment today.** Mainstream on the extensive margin, though not universal: Stack Overflow's 2025 survey found 84% of professional developers using or planning to use AI tools, with 51% using them daily. As our companion report establishes, all forecast-relevant information now lives in the *intensity distribution* — heavy users consume on the order of ten times the tokens of average users, and the growth engine is the migration of the distribution's broad middle toward codified vanguard practice.

**Trajectory and assumptions.** Our model assigns a base of 40 million developers, 1,500 tasks per year, 500K tokens per task, and an intensity-equivalent penetration rising 55% → 65% → 72%. Three definitional notes ensure consistency with the companion report. First, "penetration" here is **not adoption** (which is already broadly diffused); it is the share of the global developer base operating at the reference intensity of 750 million tokens per year — a normalization of the intensity distribution whose rise *is* the body migration. The same term carries this report's geographic scope for this category: China-domiciled developers remain inside the published global denominator, since no published source separates them; the penetration coefficient is calibrated to absorb that scope difference rather than applied on top of an explicit geographic deduction, consistent with the exclusion stated in Section 2.3. The path should therefore be read as a share of the global population reached only by developers within scope, and no separate China adjustment appears in the workbook. The 2026 figure of 55% is where the agentic-coding surge of 2025–26 lands that distribution; the two subsequent steps, to 65% and 72%, are progressively smaller as the migration approaches the ceiling set by developers whose work does not reward high-intensity delegation. Second, both the 500K tokens-per-task coefficient and the 750-million reference intensity are held flat throughout, and deliberately so. Generational loop-count collapse (iteration counts falling 4–10x per model generation as search internalizes into weights) is deflationary per task; lengthening autonomous task horizons are often cited as the inflationary offset, but a task horizon that doubles consolidates two delegations into one rather than doubling the tokens consumed — it reduces the human's coordination cost, not the agent's work — so we treat horizon growth as neutral for total volume. The 500K figure is therefore a fixed reference intensity per task rather than a forecast of the average task, and the penetration term carries only what it can carry: the migration of the developer population toward higher-intensity agentic use, which is the mechanism behind the observed rise in tokens per engineer. Observed per-task token deflation is deliberately abstracted away rather than modeled — a simplification that would understate demand if deflation slows and overstate it if deflation outruns the reallocation of agents to harder tasks. Third, the 0.70 compute weight — the coefficient that, given this category's scale, does more than any other single input to set the weighted total — reflects that coding is not a monolithically frontier workload: the autonomous-loop traffic that dominates its token count routes heavily to mid-tier and budget models — the OpenRouter token-share leadership of Chinese budget models noted in Section 3.3 is concentrated precisely in agentic coding and batch traffic — so the category runs materially below the frontier reference even though its interactive, judgment-heavy segment remains strongly frontier-selected. The category yields

16,500 trillion tokens in 2026, 19,500 trillion in 2027 and 21,600 trillion in 2028 — growth of 1.18x and then 1.11x, against a 2026 base that is itself the product of the steepest acceleration this category has recorded. It remains the dominant workload, its share declining from 82% to 57% as the rest of the economy scales but never surrendering the lead, because it remains the function with the deepest agentic penetration. What changes is its role: from 2027 it is the largest workload but no longer the fastest-growing one.

### 4.2 Quantitative Analysis & Data Science — two waves: the ungated expert core and the semantic-layer gate

**Definition and boundaries.** Work that uses code as a disposable instrument to produce a numerical answer: backtests, cohort and funnel analyses, demand forecasts, risk models, econometric studies, BI reporting. The boundary with Category 1 is the deliverable — reusable code there, an analytical conclusion here — which is why data science, whose output is a fitted model or a quantified finding, sits here rather than in Category 1 despite sharing the coding toolchain; the high-intensity coding data scientists are simply this category's earliest adopters.

**Work characteristics and verification.** Semi-deterministic: execution results and statistical diagnostics verify mechanically; a residual human judgment closes the loop. Crucially, the harness is inherited wholesale from coding — which means this category is pre-equipped for autonomous iteration the moment its other constraint falls.

**Two adoption paths, one gated and one not.** Generating a SQL query or a cleaning script is token-light to specify but token-heavy to get right on messy, non-standard data — and data preparation, not modeling, is where most analytical time and most agent iterations go. Two distinct populations drive this category on two different schedules. The first is today's expert core — dedicated analysts, data scientists, and accountants who already hold the business semantics in their heads and can therefore direct an agent precisely by prompt, with no explicit ontology required. For them the enabling event has already happened: tool-embedded agents now sit inside the applications where their work lives — Claude for Excel reached general availability across all paid plans in 2026, notebook assistants (Gemini in Colab and equivalents) automate cleaning and analysis end to end — and these run against local spreadsheets and notebooks without any enterprise semantic layer. The second population is the far larger body of finance, operations, and marketing professionals who do *not* hold the semantics and cannot self-serve until the data is made legible to them. The distinction between the two paths is a distinction between two kinds of data, and it is why the semantic-layer gate binds less than a naive reading suggests: analysis run against files the user already holds — a spreadsheet, an export, a notebook — needs no enterprise ontology, and it is this tool-embedded, file-level segment that the add-ins now in general availability address; the gate applies to warehouse-level analysis against governed data. Both paths share the property that makes this category the fastest-diffusing non-coding function: its core feedback loop is machine-checkable. Code executes or fails; a reconciliation ties or does not; a model's diagnostics are numbers. That is what lets an agent iterate unattended through most of a task — the residual human judgment that closes the loop arrives at the end rather than at every step — and it is also what makes the felt utility to a non-expert so large, since an analysis the user could not have performed at all is delivered with its own evidence of correctness. For them the gating investment is the **enterprise semantic layer** — metadata catalogs, ontologies, certified metric definitions

— which we expect to become a defining enterprise-software theme of H2 2026, executed substantially by forward-deployed engineers (the Palantir FDE model now being generalized across the industry) and platformized by the data-stack incumbents (Databricks Unity Catalog, Snowflake, dbt, Palantir's ontology). In the language of our earlier work, the semantic layer is a control-plane asset: whoever owns it owns the gateway through which the second, larger wave of this category's token demand will flow.

**Trajectory and assumptions.** The base is the analysis-capable professional population, anchored at 178 million by scaling Microsoft 365's ~450 million commercial paid seats by the US BLS share of business-and-financial-operations plus management occupations (39.6%). Penetration follows 10% → 24% → 42%. The 2026 figure (10%, ~18 million effective users) reflects the expert core already running agentic cleaning and analysis through tool-embedded add-ins — the great majority of the ~24 million dedicated analysts and data scientists, a level consistent with Excel-add-in general availability and its spread to accountants. The 2027 figure (24%) and 2028 figure (42%, ~75 million) add the second wave as the semantic-layer build-out lets non-expert professionals self-serve; the curve is steeper than for any other non-coding function because machine-checkable feedback and large felt utility are the two conditions under which agentic tools have diffused fastest, and this is the only non-coding function that has both. Full democratization still lies beyond the horizon. The activity rate is 450 tasks per year — roughly two analyses per working day for a penetrated professional, the frequency appropriate to a tool that sits inside the spreadsheet rather than beside it. Tokens per task are held at 250K — a level that does not distinguish cleaning from modeling, since data preparation is at least as cognitively demanding and iteration-heavy as causal analysis, and the metric here is code-*generation* tokens rather than the CPU load of code execution. With a compute weight of 0.6, the category grows from 2,003 trillion tokens (2026) to 8,411 trillion (2028) — a 4.2x expansion that makes it the clear second workload behind coding and, by 2028, the largest post-coding function by a factor of more than two, and the principal inheritor of the growth that coding relinquishes. The dominant uncertainty is the pace at which the second, gated wave arrives.

### 4.3 Transaction Record Operations — the deterministic long tail beyond RPA

**Definition and boundaries.** The per-item processing of structured business records flowing through existing systems: bank and account reconciliation, month-end close, accounts-payable and receivable invoice handling, KYC/AML screening, insurance claims adjudication, order and payroll processing, and contract screening against standard clauses. This is *not* database engineering (Category 1 builds the pipes; this category processes what flows through them) and not judgment work (detecting a non-standard indemnity clause is rule-checkable and belongs here; deciding whether the clause hurts this particular deal belongs in Category 6). It corresponds to the substance of the Office & Administrative Support occupational group whose usage, uniquely in the Economic Index data, rose on both consumer and API surfaces — with billing- and payment-support tasks among the most prevalent automated API workloads.

**Work characteristics and verification.** The rare and strategically decisive combination: document- and text-based work whose verification is nevertheless deterministic — the numbers tie or they do not, the record matches or it does not. This is the only text-heavy function with a compiler-equivalent, which means autonomous retry loops work, and full automation (rather than augmentation) is the natural endpoint.

**Deployment today and the legacy baseline.** Adoption is accelerating from the supply side — Anthropic's May 2026 financial-services launch made ready-to-run agent templates for reconciliation, month-end close, and KYC screening generally available, alongside citations of broad Claude deployments across banks, asset managers, and insurers (though without production-deployment counts for the newly released templates) — and from the demand side, with surveyed agentic-AI use among finance teams at 6% and a further 38% planning adoption within a year. But the addressable base is narrower than the gross: structured high-volume matching was largely conquered by RPA and rules engines years ago. The LLM increment is the unstructured residual — exception queues, free-text remittance advice, non-standard documents — sized here at roughly 60 billion items per year, against gross transaction flows several times larger.

**Trajectory and assumptions.** 60 billion addressable items, 40K tokens per item, penetration 3% → 10% → 22%, compute weight 0.10 — budget models are decisively sufficient here, and the workload's batch-parallel, latency-tolerant character gives it the lightest compute profile in the taxonomy. The low 2026 start reflects that although verification is deterministic, autonomous adoption on the exception residual is still early. Tokens grow from 72 trillion (2026) to 528 trillion (2028). The figure understates the category's strategic importance: because volume scales with business activity rather than headcount, and because unit costs are collapsing toward zero, this is the category most exposed to the Jevons upside scenario of Section 6.

### 4.4 Sales & Outreach — parallel research at near-zero marginal cost

**Definition and boundaries.** Outbound, asynchronous sales work initiated by the firm: account research, lead qualification, CRM enrichment, personalized outreach, proposal drafting. Two tests separate it from Category 5: direction (the firm initiates, rather than the customer) and timing (asynchronous batch — the work can run overnight, rather than a live exchange). The counterparty is not a test: an upsell campaign to existing customers is classified here, because workload properties, not the org chart or the recipient's status, drive the taxonomy. Category 5 is correspondingly the customer-initiated, synchronous side of the same boundary.

**Work characteristics and verification.** Outcome-verified (reply and conversion rates) and batch-parallel. The unit of work — and this is essential to reading our token coefficient — is not "one email" but **one multi-step research package per lead**: the agent browses the prospect's site, LinkedIn, news, and funding history, each fetch depositing thousands of tokens of page content into context; the email is the small terminal output of a deep-research process. Because no human sits between instances, ten thousand leads means ten thousand parallel agents — the first mass-scale demonstration that parallelization, not per-task depth, can drive aggregate token demand.

**Deployment today.** This was one of two workflow families the Economic Index flagged as having more than doubled in API share within three months (sales-collateral generation, B2B lead validation, cold-email composition) — growth from a small base, as legacy marketing-automation platforms already owned templated mass-mailing.

**Trajectory and assumptions.** A base of 40 million (Microsoft 365's ~450 million seats scaled by the US BLS share of office-based sales plus the outreach half of marketing and PR occupations, ~8.9%), 1,000 lead-packages per year, 100K tokens per package, penetration 2% → 8% → 20%, compute weight 0.15 (DeepSeek-class models suffice). The low start reflects that today's use

is mostly draft-generation augmentation rather than autonomous research batches. Tokens grow from 80 trillion to 800 trillion. As with Category 3, the modeled path excludes the volume-elasticity upside: when researching a lead costs cents, the rational response is to research far more leads.

### 4.5 Customer Interaction — the linear baseload and the native-multimodal shift

**Definition and boundaries.** Inbound, real-time service of existing customers: billing inquiries, refunds, technical support, retention. The mirror image of Category 4 on both workload tests — customer-initiated rather than firm-initiated, and synchronous, with millisecond latency obligations, rather than asynchronous. An existing relationship is the typical setting here rather than a test of the boundary.

**Work characteristics and verification.** Outcome-verified at the population level (resolution rates, CSAT) and executed on a real-time streaming harness. Per-conversation tokens are bounded and loops are short, so growth is linear in interaction volume rather than amplified per task — a utility-like baseload rather than a compounding curve.

**Deployment today.** The most commercially mature post-coding category. Sierra scaled from $26 million ARR at end-2024 to roughly $200 million by May 2026, with voice overtaking text as the primary channel and call volumes in the hundreds of millions annually; Decagon reached a $4.5 billion valuation; pricing across the category has migrated from seats to outcomes (roughly $1–1.5 per resolution). Vendor-reported deflection rates of 70–80% against independent enterprise medians nearer 41% suggest the production reality is strong but less uniform than marketing implies.

**Trajectory and assumptions.** 80 billion interactions per year, 20K tokens per interaction, penetration 3% → 12% → 28%, compute weight 0.20. The 20K coefficient is a single blended figure held across all three years, not a 2026 reading: it assumes a mix in which native speech-to-speech already handles a substantial share of traffic. For tractability the coefficient is held flat rather than modeling the native-speech mix separately by year, which may make it generous early in the forecast, while native pipelines are still a minority of traffic, and conservative late, as they become the majority; Section 7 tracks that mix directly. The weight sits above outcome-verified Category 4's 0.15 for a per-instance-risk reason: an outbound email that misses simply dissolves into the A/B statistic, while a wrong answer here lands on a live customer, so deployments avoid the very cheapest tier — even as latency SLAs forbid the frontier one and cap the weight far below the judgment categories. The token coefficient embeds an architectural forecast worth making explicit: today's dominant stack splits the pipeline (ASR → text LLM → TTS), leaving the LLM only ~8K text tokens per conversation; as native speech-to-speech multimodal models — which ingest and emit audio tokens directly at hundreds to a thousand tokens per minute — take share, audio becomes LLM tokens, and our 20K figure is the blend. Tokens grow from 48 trillion to 448 trillion. The penetration path recognizes that although CCaaS platforms (Genesys, NICE, Five9) are commercially mature and vendors like Sierra and Decagon report real deflection, truly autonomous resolution remains a minority of total interactions; commercial maturity nevertheless supports the steepest penetration ramp among the volume-based categories (3 → 28%, against 3 → 22% for transaction operations). One infrastructure-layer note carries forward to the companion volume: latency SLAs cap serving

batch sizes, so this category's effective hardware demand runs above what its token count implies.

### 4.6 Qualitative Text Review & Deep Research — process tokens versus the reading-speed cap

**Definition and boundaries.** Work whose deliverable is argued judgment over unstructured text: equity research, legal due diligence and discovery review, M&A strategy memos, policy analysis, literature review. The boundary with Category 3 is structure: screening contracts for deviant clauses is rule-checkable (Category 3); deciding whether the deviation is acceptable in this deal is judgment (here).

**Work characteristics and verification.** The cognitive-serial regime: the only verifier is a human reading at 200–300 words per minute, under a hallucination tax that in legal and audit contexts is catastrophic per error. This is the hardest verification problem in the taxonomy, and the category's penetration curve is gated less by model capability than by **verification UX** — data lineage, citation mapping, and critic-agent frameworks that compress the human review window.

**Why tokens grow anyway.** Two mechanisms decouple token demand from reading speed. First, **process tokens**: a deep-research agent's final 10K-token memo is the residue of hundreds of thousands of tokens of autonomous searching, fetching, and cross-checking — consumption that occurs regardless of how fast anyone reads the output. Second, **parallelization**: reading speed throttles the review of each artifact, not the number of agents a professional runs concurrently; a partner queuing ten diligence agents multiplies burst demand ten-fold at unchanged reading speed. The reading-speed cap is real, but it binds sustained throughput — how many runs a professional commissions over a year — rather than the concurrency of any single burst or the tokens each run consumes.

**Why growth is nevertheless capped: the inverted effort ratio.** The two mechanisms above decouple tokens per run from reading speed; a third mechanism recouples run *frequency* to it. Unlike Categories 1 and 2 — where an agent can work unattended for days and the human verifies the outcome in minutes, because the tests pass or the deliverable is a few tables and charts — a qualitative agent working for thirty minutes returns a document the professional must then spend hours reading closely, under a hallucination tax that makes skimming reckless. The AI-to-human effort ratio is the precise inverse of the coding case, so the scarce resource is not agent capacity but the commissioning professional's review bandwidth: there is no point launching research one cannot read, and unread output has no value. The market already prices this in — OpenAI's published quotas allow 25 deep-research queries per month on Plus, Team, Enterprise, and Edu plans and 250 on Pro, with current limits varying by plan — and the fact that such caps bind for heavy users is direct evidence that per-user run frequency, not model capability, is the binding constraint on this category's volume.

**A qualification on the cap, from the power-user tail.** The review-bandwidth constraint is a constraint on human reading, and a minority of practitioners have begun to route around it: rather than read a long artifact, they have two models critique and revise each other's output over several rounds, read only the critiques, and read the full document once the models converge. In that workflow the human verifies a disagreement log rather than the artifact, and the effective verifier becomes partly a machine. This is the mechanism by which the category

could eventually escape its cap, and it is worth watching for that reason. It remains confined to a small tail of research-intensive users, however; for the population as a whole one to three revision passes are the norm, so its effect on demand is a modest rise in tokens per run rather than a lifting of the frequency ceiling.

**Trajectory and assumptions.** A base of 96 million — Microsoft 365's ~450 million seats scaled by the US BLS share of business-professional judgment occupations plus research occupations (21.3%, after excluding structured-processing lines and adding life/physical/social-science researchers on both sides of the ratio) — at 150 tasks per year and tokens per task rising from 400K (2026) to 500K (2027) and 600K (2028). The token coefficient covers deep-research process tokens and long-document prefill (ingesting a five-hundred-page report alone approaches 300K tokens); its rise of roughly 20–25% a year reflects the gradual spread of machine cross-review, in which each review round re-ingests the document. The frequency coefficient is held flat at 150 runs per year, roughly 12–13 per month, one substantive review or research run every 1.7 business days, against published plan quotas of 25–250 per month. Two forces cancel here: incumbent users run more as verification UX improves, but the professionals who enter the penetrated base later are, on average, lower-frequency users than those already in it, so the population mean does not rise. Penetration is 18% → 26% → 35% — the highest 2026 start of any non-coding function and the flattest climb. Both properties follow from the same fact: deep research is the easiest agentic capability to reach (it ships across all three major assistants plus Perplexity and Genspark, requires no integration, and produces a familiar artifact), so a large share of the base is already using it intermittently; but it is a mature, human-verified workload whose ceiling is set by reading bandwidth, so the climb from there is moderate. Compute weight is 0.80, the highest in the model, consistent with the Economic Index finding that higher-wage tasks select frontier models most strongly. Tokens grow from 1,037 trillion to 3,024 trillion — a 2.9x expansion, the slowest of any non-coding function, that nonetheless keeps this the third-largest category in nominal terms and the second-largest post-coding contributor on a compute-weighted basis. Its climb is governed by verification-UX maturity, and its ceiling by human reading bandwidth, rather than by model capability. One silicon note for the companion volume: as cross-review spreads, the category's token mix shifts toward prefill, which is compute-bound, and away from the memory-bandwidth-bound decode profile of the autonomous-loop functions.

## 4.7 Business Workflow Integration & Automation — the control-plane-gated frontier

**Definition and boundaries.** Cross-system automation that orchestrates actions across multiple enterprise systems — CRM, ERP, project trackers, mail, ticketing — through text interfaces: APIs, MCP servers, and CLIs. The canonical unit is not a single assistant convenience (a drafted email, a meeting summary) but a multi-system workflow: *contract signed → update CRM → generate invoice → create project tasks → notify stakeholders*. This is a workflow-orchestration function, not a clerical one: personal-assistant conveniences are its shallow end, while enterprise workflow orchestration across systems is its substance and the source of its token intensity.

**Execution is text, not screenshots.** This is not a screenshot-driven, computer-use workload. Pixel-level GUI agents are heavily demonstrated but appear far less prevalent in production than connector-based execution, constrained by latency, cost, reliability, and security; the

execution path that ships is structured connectors — MCP, Graph API, per-application REST — which is how, for instance, Microsoft 365 Copilot integrates. Because execution is text rather than high-cost screenshot tokens, the workload caches well and carries a light per-token compute profile. The 60K tokens-per-task coefficient is a blended average across a task mix that the ~2-tasks-per-business-day frequency implies: the majority are shallow single-connector actions (10–20K tokens), with a minority of deep multi-system chains running to 150–250K — each step of a chain re-injects tool schemas (MCP tool definitions alone run to hundreds or thousands of tokens per call, cached but still counted on the billed-token basis of this model), carries intermediate results back into context, and passes approval turns. This multi-step chain unit, not Category 3's single-record unit (one invoice exception, one reconciliation break, resolved within the context of a single document and rule), is why the coefficient exceeds Category 3's 40K even though each individual step is lighter.

**Work characteristics and verification.** The binding constraint is **irreversibility without a verifier**. Sending an email, approving a payment, and mutating a record are side-effectful actions, and — unlike Category 3's reconciliations or Category 1's compiles — there is no deterministic oracle to confirm the action was correct before its consequences land. With no machine verifier, autonomous loops cannot close; human-approval gates persist; and the category is structurally confined to augmentation rather than full automation. This is precisely why its penetration, though its ceiling is high, climbs more conservatively than Category 2's — the semi-deterministically verified function and the unverified one share a gate but not a verifier, and the verifier decides the pace. The same property sets the compute weight, and explains why it sits at 0.25 — above outcome-verified Category 4's 0.15 despite the lighter per-step content. Category 4's failures are cheap and statistically verified: a weak email costs nothing per instance, and the A/B loop tolerates errors, so the cheapest sufficient model wins. Category 7's failures land on irreversible actions with no verifier, so the error budget must be spent on first-pass reliability — and multi-step reliability compounds ruthlessly (95% per-step accuracy over an eight-step chain completes only 66% of workflows; 99% completes 92%), while tool-calling and instruction-following accuracy is precisely where budget models degrade most against frontier models. Production orchestration stacks (Microsoft 365 Copilot, Agentforce) accordingly run on mid-to-frontier models rather than the token-cheapest tier.

**The shared gate with Category 2.** This category and Quantitative Analysis are a pair, though not symmetrically: Category 2's *second* wave and the bulk of this category are unlocked by the same prerequisite — the enterprise semantic ontology and control plane — while Category 2's first wave, running on files the user already holds, is not gated at all. Where Category 2 *reads* the governed data layer to produce analysis, this category *writes and acts* across the systems that layer connects. Both therefore inflect together in 2027 once the metadata and ontology build-out of H2 2026 matures, and both are control-plane-gated in the precise sense of our earlier work — which makes this, on the semantic-layer thesis, one of the most strategically important growth frontiers in the taxonomy, and a central focus of this research series.

**Trajectory and assumptions.** The base is Microsoft 365's ~450 million commercial paid seats (FY26 Q2 disclosure) — the single most auditable anchor in the model, representing the office-software installed base this function acts across; Google Workspace, reported as ~11 million paying *organizations* — an organization count, not a seat count, and therefore not directly comparable — is treated as unquantified upside rather than added to the base. At 500 tasks per year,

60K tokens per task (the blended shallow/deep mix above), penetration a conservative 3% → 11% → 24%, and compute weight 0.25, tokens grow from 405 trillion to 3,240 trillion — an eight-fold expansion, large in nominal terms but, at the lowered weight, a more modest contributor to compute-weighted demand. The curve is held below Category 2's throughout despite the shared gate, though production MCP-server and connector deployments have moved cross-system automation from demonstration into routine use faster than the verification problem alone would have predicted. The conservative penetration is the direct consequence of the verification problem above: this function shares Category 2's semantic-layer gate but lacks its semi-deterministic verification loop — there is no machine-checkable signal to close on before an irreversible action lands — so human-approval gates persist and adoption trails. Realized penetration depends on control-plane maturation and on trust engineering (safe-action sandboxing, approval UX) whose schedule is among the least forecastable elements in the model.

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# 5. The Quantitative Demand Model

## 5.1 Architecture

For each of Categories 1–7, annual tokens = global base (workers, seats, or transaction volume, in millions) × annual AI tasks per unit × tokens per task × penetration. Penetration is throughout an *effective*, intensity-normalized rate rather than a binary adoption count, but it denotes two different things depending on how the base is built. For the workforce-based categories it is the share of the base operating at the category's reference task frequency and token intensity; Section 4.1 develops that distinction explicitly for the category where the gap between an effective and an adoption reading is widest. For the volume-based Categories 3 and 5 it is the share of addressable items or interactions processed agentically at the reference token intensity. The workforce bases are built on an installed-base rather than a demographic method, but by two different routes. The non-coding workforce categories (2, 4, 6 and 7) are anchored to Microsoft 365's ~450 million commercial paid seats and partitioned with US BLS occupational ratios, so populations without a paid seat are excluded by construction. Category 1 is anchored differently: its 40 million is taken from published global developer-population estimates for the reasons set out in Section 4.1, with GitHub's ~150 million registrations serving only as an outer-TAM check and no occupational partition applied. Categories 3 and 5 are volume-based and use neither route. The full derivation, with a confidence grade on every input, is in the Base Sizing sheet of the companion workbook. The grades run A to D on a single axis — how directly the figure is observed: A, a directly observed or audited input (a disclosed seat count, a single-occupation actual); B, a whole occupational major group or an explicit published concept; C, a structured judgment or arbitrary split where no published basis exists; D, a forecast judgment. Source quality and allocation judgment are graded on the same scale, so an auditable seat count partitioned by a grade-C ratio yields a grade-C base. The model is intentionally free of prefill/decode decomposition, cache adjustment, and intra-year dynamics: it is an order-of-magnitude instrument whose every input is exposed for revision. A per-category compute weight (frontier = 1.0) converts nominal tokens into frontier-equivalent tokens — the series the companion infrastructure volume takes as its demand input.

## 5.2 Assumption anchors

Each coefficient is disciplined against an external anchor documented in the model workbook: the OpenRouter finding that programming exceeded half of routed tokens as of the platform's late-2025 usage study (Category 1's dominance); the Economic Index's doubling of sales-automation and trading-operations workflows (Categories 3–4 momentum) and its wage-to-Opus gradient (the compute weights); Sierra's disclosed call volumes (Category 5); published deep-research quotas — 25 queries per month on OpenAI's Plus, Team, Enterprise, and Edu plans and 250 on Pro — as the upper bound beneath which Category 6's 150-runs-per-year frequency is set; the per-developer intensity distribution from the companion report (Category 1's reference intensity of 750M tokens per developer-year sits in the upper-middle of the observed distribution, appropriate for a pool defined to exclude light users); the installed-base anchors for the non-coding workforce categories (Microsoft 365's ~450 million commercial seats scaled by US BLS occupational ratios, documented line by line in the Base Sizing sheet) and the published developer-population estimates behind Category 1's base; and — as validation only, never as a basis for adjusting any coefficient — a set of external reference points against which the total is checked for plausibility. Our 2026 total of ~20,100 trillion tokens per year (~1,680 trillion per month) compares to OpenRouter alone at a reported ~100 trillion per month; to Google's disclosed throughput of more than 3,200 trillion tokens per month as of May 2026 (a seven-fold rise year-over-year, spanning consumer, Search, and internal surfaces), within which Google's model APIs alone run at roughly 19 billion tokens per minute, on the order of 830 trillion per month; and to the token volume implied by Anthropic's disclosed run-rate revenue, roughly 80% of which is business revenue, which at plausible blended prices for cache-heavy agentic traffic corresponds to several hundred trillion tokens per month from one vendor. A B2B-only, all-provider, seven-function estimate of ~1,680 trillion per month sits inside that corridor. The check runs in one direction. It can establish that the bottom-up total is of the right order; it cannot set any input, and by construction does not. Every figure in Table 2 is the product of the category-level coefficients in Section 4.

## 5.3 Outputs

**Table 2. Annual B2B LLM token demand by category (trillions)**

| Category | 2026E | 2027E | 2028E | Share '26 | Share '28 | Growth '26→'28 |
|---|---|---|---|---|---|---|
| 1. Software Engineering | 16,500 | 19,500 | 21,600 | 81.9% | 56.8% | 1.3x |
| 2. Quantitative Analysis & Data Science | 2,003 | 4,806 | 8,411 | 9.9% | 22.1% | 4.2x |
| 3. Transaction Record Ops | 72 | 240 | 528 | 0.4% | 1.4% | 7.3x |
| 4. Sales & Outreach | 80 | 320 | 800 | 0.4% | 2.1% | 10.0x |
| 5. Customer Interaction | 48 | 192 | 448 | 0.2% | 1.2% | 9.3x |
| 6. Qualitative Review & Research | 1,037 | 1,872 | 3,024 | 5.1% | 7.9% | 2.9x |
| 7. Business Workflow Integration & Automation | 405 | 1,485 | 3,240 | 2.0% | 8.5% | 8.0x |

| Category | 2026E | 2027E | 2028E | Share ’26 | Share ’28 | Growth ’26→’28 |
|---|---|---|---|---|---|---|
| **Total** | **20,144** | **28,415** | **38,051** | 100% | 100% | 1.9x |
| **Compute-weighted total (frontier-equivalent)** | **13,711** | **18,513** | **23,658** | | | 1.7x |

*Category rows are rounded independently; totals are computed on unrounded model output and may therefore differ from the sum of the displayed rows by one unit.*

Four readings of the table. First, coding’s dominance remains decisive but no longer near-total: it holds 82% of nominal tokens in 2026 and 57% in 2028 — and, because it carries a 0.70 weight against the budget-model categories’ 0.10–0.25, a still-larger share of compute-weighted demand, roughly 84% in 2026 easing to 64% by 2028. The table’s most important feature is the shape of that decline. Coding’s 2026 base is the product of the steepest acceleration the category will see; its year-over-year growth then falls to 1.18x and 1.11x, so that of the roughly 17,900 trillion tokens the seven functions add between 2026 and 2028, coding contributes 5,100 trillion and the six post-coding functions 12,800 trillion. Coding remains the largest workload throughout; from 2027 it is no longer the growth engine. Second, the growth column tracks the verification framework, with one instructive exception. The functions released from serial human review — reconciliation, verified by arithmetic; sales research and customer interaction, verified statistically by outcomes; and above all quantitative analysis, verified by execution — grow four- to ten-fold. The function whose verifier is a human reading grows 2.9x, and the function that had already saturated its verifier grows 1.3x. The exception is workflow automation, which grows eight-fold despite having no verifier at the moment of action: its expansion is catch-up from a 3% base as the semantic layer and control plane it depends on are built, not a graduation to autonomy, and its 24% endpoint remains among the lowest in the model — the verification constraint shows up in the level it reaches, not in the rate at which it closes the gap. That ordering emerges from the category-level assumptions rather than being imposed as an explicit growth-rate constraint, and it functions as an internal consistency check on the framework of Section 3 rather than as independent evidence for it — the penetration paths are themselves placed on the verification axis, as Section 6 states. Third, by absolute 2028 token contribution the post-coding order is Quantitative Analysis & Data Science (8,411T) > Business Workflow Integration (3,240T) > Qualitative Research (3,024T) — the last two nearly tied in nominal terms but far apart on a compute-weighted basis (2,419T weighted for Qualitative Research against 810T for Workflow Integration), a within-table illustration of the token-versus-compute principle: the connector workload emits more tokens, the judgment workload consumes more compute. Quantitative Analysis leads the post-coding field on both bases throughout, by a widening margin. Fourth, under the assumed weights the weighted total runs 32–38% below the nominal total — the quantitative expression, within this model, of the principle that token statistics read naively overstate infrastructure demand; the margin is a scenario output of the weight assumptions, not a measured error band. The divergence widens over the forecast precisely because the fastest-growing categories carry the lightest weights: Categories 3–5 and 7 together reach 13% of nominal tokens by 2028 while contributing under 5% of compute-weighted demand, and Category 6 does the reverse. The

frontier-equivalent series in the table's final row, not the nominal total above it, is the figure the companion infrastructure volume converts into hardware requirements.

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## 6. Risks and Sensitivities

**The Jevons scenario (upside, Categories 3 and 4, with a related channel in Category 5).** The addressable bases for Categories 3 and 4 are sized to the legacy-automation residual at current unit costs; Category 5's base is an interaction count rather than a residual, but it is subject to the same elasticity through the volume of interactions it becomes economic to handle agentically. But budget-model pricing is approaching zero marginal cost, and the rational enterprise response is to expand the perimeter — re-platforming work currently handled by RPA, and applying LLM treatment to volumes never before economic. In that "low-density, high-volume" scenario, nominal token totals exceed our forecast while the compute-weighted total grows far less — a divergence whose infrastructure consequences the companion volume takes up.

**Semantic-layer timing (the second-wave gate).** Category 2's near-term growth rests on tool-embedded adoption by the expert core and is not gated; its second, larger wave — non-expert self-serve analysis, and the bulk of Category 7's workflow automation — is conditional on enterprise metadata infrastructure being built on schedule. That build-out is services-heavy, organizationally invasive, and historically prone to slippage; each quarter of delay defers the second wave for both categories. The leading indicators of Section 7 exist largely to monitor this gate.

**Base sizing and its confidence structure.** Category 1's base is the single largest lever in the model. Its 40 million sits at the upper band of published developer counts for the reason given in Section 4.1; at the narrower 30 million professional-developer figure every coding row in Table 2 falls by a quarter, the most consequential single sensitivity in this report. The workforce bases for the other categories are anchored to installed-base figures scaled by US BLS occupational ratios: Quantitative Analysis to 178 million (business-financial + management occupations, 39.6% of the office-software workforce), Sales & Outreach to 40 million (office-based sales + outreach share of marketing/PR, 8.9%), Qualitative Review to 96 million (business-professional judgment + research occupations, 21.3%), and Business Workflow Integration to Microsoft 365's ~450 million seats directly. These carry heterogeneous confidence on the single A-to-D scale of Section 5.1: the seat anchors are directly disclosed (grade A), the major-group ratios rest on published occupational groups (grade B), a few line-level splits are arbitrary 50:50 allocations where no basis exists (grade C, low sensitivity — under 8% of Category 2 and 17% of Category 4), and the volume categories (3 and 5) rest on transaction-count estimates with the widest bands (grade D). Two structural caveats attach: all ratios reflect US labor structure applied to a global seat base (defensible because paid seats already concentrate in developed-economy enterprises, but a stated assumption), and populations overlap across categories — the same professional is counted under analysis, judgment, and workflow functions because they perform all three, which is valid in a functional model but must not be summed into a headcount.

**Penetration is forecast, not measurement.** Every penetration rate except Category 1's is a principle-based judgment on the verification-mechanism axis (grade D), not an observed figure. Category 1 alone has an empirical anchor — not the adoption statistic (84% of professional developers using or planning to use AI tools, a figure that is already near-universal and, per Section 4.1, is not the quantity being forecast) but the per-developer intensity distribution documented in the companion report, against which the intensity-equivalent penetration is normalized; Category 6 has an indirect one (Deep Research triangulation); the rest are placed by the logic that machine-verifiable functions are the ones able to follow coding up the autonomy curve, which is why the non-coding functions start at 2–18% in 2026 rather than the higher figures a naive reading of vendor adoption claims would suggest, and end the forecast spread across 20–42%. What the verification axis orders is not those endpoints — which reflect starting levels and gating infrastructure as much as verification — but the growth trajectories built on them, as Section 5.3 sets out. The single largest swing factor is the timing of the semantic-layer build-out, which gates both Category 2 and Category 7; a one-year slip shifts those curves right by roughly a year.

**Coefficient sensitivity to definitions.** Tokens-per-task coefficients are hostage to two specifications — the unit of work and the architecture assumed. The clearest cases: Category 5's coefficient depends on whether voice runs as a split ASR-LLM-TTS pipeline or a native speech-to-speech model; Category 4's 100K is a per-lead research package, not a per-email figure; Category 7's 60K is a blended average over a mostly-shallow task mix whose deep-chain tail does the token work; and Category 6's pairing — 400–600K per task at a flat 150 tasks per year — lets the per-run coefficient rise with machine cross-review while bounding frequency by review bandwidth, where the alternative decomposition (more, lighter tasks) yields a similar product but a different falsification target. Category 6's frequency is the one coefficient outside Category 1 anchored to a directly published figure (platform rate limits) rather than disciplined against an indirect consistency check, which upgrades it from pure judgment. Any forecast that does not state these specifications should be distrusted.

**Measurement caveats.** Figures are billed-token basis including cache reads; the scope exclusions of Section 2.3 apply throughout (chatbot-tier work, personal free-tier use, China-domiciled demand, R&D and other agentic domains without a base population); occupational statistics will progressively misclassify non-coding work executed through coding harnesses (Section 2.1); and vendor run-rate revenue figures carry documented internal tensions and deserve a discount, as the companion report details.

**Task-horizon inflation (the principal upside case for coding).** Autonomous task horizons are lengthening rapidly, and a common reading treats that as inflationary for tokens per task. Section 4.1 argues the opposite: a longer horizon consolidates two delegations into one rather than doubling the work, so the coefficient is held flat. Readers who take the inflationary view can scale the coding rows directly — each 20% rise in tokens per task adds roughly 4,300 trillion tokens to the 2028 total — though the resulting trajectory implies a coding deceleration materially shallower than the one the rest of the evidence supports.

**Correlated-verification risk.** The entire amplification thesis for Categories 1–3 rests on machine-checkable verification signals being trustworthy — the compiler and test suite in Category 1, execution results and statistical diagnostics in Category 2, the reconciliation itself in Category 3. A high-profile correlated failure of model-based verification layers in production

would re-impose human review and compress the autonomous-loop categories toward their pre-agentic token profiles.

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## 7. Tracking Indicators

The model's value lies in being falsifiable on a short horizon. We track, by category: **(Categories 2 and 7, shared gate)** data-catalog and semantic-layer bookings at Databricks, Snowflake, dbt and Palantir, and FDE and analytics-engineer hiring volumes — the leading indicator for *both* the read-side (Category 2 analysis) and write-side (Category 7 workflow automation) inflections, since both are gated by the same ontology build-out; plus, for Category 2 specifically, the appearance of execute-and-verify text2SQL agents (query → error → revise loops) in production stacks — the cleanest signal that the lighter, query-drafting end of its task mix is migrating from the tens of thousands of tokens it consumes today toward the hundreds of thousands that iterative cleaning-and-analysis work already burns, the migration on which the model's blended 250K coefficient rests; and, for Category 7 specifically, production MCP-server and connector deployment counts as the proxy for cross-system automation moving from demo to production. **(Category 1)** the indicator suite of the companion report, led by the full-delegation rate and within-firm top-decile token share — and, as the direct test of this report's central claim, the year-over-year growth of coding token volume, which the model expects to fall below 1.2x by 2027. **(Categories 3, 6)** adoption telemetry of financial-services agent templates (reconciliation, close, KYC) and the maturation of verification-UX features — lineage, citation mapping — in research tooling. **(Category 5)** the share of production voice stacks running native speech-to-speech models rather than split pipelines. **(Cross-cutting)** the token share of Chinese budget models in agentic workloads on OpenRouter — the live read on our compute weights — and the spread between vendor dollar-revenue growth and token-volume growth.

A final synthesis. The 2026–2028 period will be remembered, we believe, as the interval in which the question "which industries adopt AI" gave way to "which verification mechanisms automate next." Coding answered first because its verifier was already a machine. The functions that follow — reconciliation because its verifier is arithmetic, quantitative analysis because its verifier is execution, research because its verifier is being engineered — will follow in the order their verification costs fall — a conceptual ordering of difficulty rather than a calendar, since in this model the largest and earliest post-coding demand comes from quantitative analysis, whose verifier is already in place. Token demand is the shadow this sequence casts on the infrastructure market; this report measures the shadow in tokens, and the companion volumes read it onto consumer media and silicon.

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*Disclaimer: This report is for informational purposes only and does not constitute investment advice. Quantitative estimates herein are order-of-magnitude modeling assumptions, not measured data, and are documented as adjustable inputs in the companion workbook (Agentic AI Token Demand Model workbook). Vendor-disclosed figures (run-rate revenues, deflection rates) carry the caveats noted in the text.*

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